



STUDENT NAME: SHEHRYAR SHEHRYAR

STUDENT ID: P2952028

MODULE TITLE: AI FOR MOBILE ROBOTS

Shehryar Shehryar
DE MONTFORT UNIVERSITY, LEICESTER

Introduction:

For an autonomous mobile robot, it needs to have the capability of perception and representation for its surroundings in order to operate safely. One of the key features that enables this is environment mapping.

This project is about developing a 2D map of an unknown environment with the use of TurtleBot3 robot and robotics operating system (ROS). The controller receives only odometry and range somewhere, readings are received and does not depend on any ROS navigation or mapping libraries.

Map Construction Technique:

I have used Occupancy Grid Mapping in which the environment is divided into two equally sized grid cells. Each cell represents the probability of being occupied by an obstacle.

➤ Sensors Input:

- The odometry (/odom) contains the estimated position and orientation of the robot.
- Laser scan (/scan): It returns distance measurement toward surrounding obstacles.

➤ Mapping Workflow

1. Read the position and orientation of a robot (x, y, θ) in odometry.
2. It reads and validates the laser range data.
3. Transform laser readings from Local to Global map coordinate.
4. Convert to grid coordinates from global.
5. Mark corresponding cells as an occupied
6. Continuously Update the map as the Robot Moves.

Software Implementation:

The system is implemented in Python and operates in ROS. Sensor readings are processed by a custom ROS node called mapper in order to build up the map.

➤ Main Functional Components:

1. **Map Class:** Stores and Publish the Grid Cells.
2. **Odometry Callback:** Get robot pose and convert quaternion to yaw angle.
3. **Laser Scan Callback:** Laserscans and Updates Grid cells.
4. **Map Publisher:** Publishes map and metadata for RViz visualisation.

As ROS supplies new sensor data all the time, mapping is performed in real-time.

Visualization and Screenshots:

In this I presents visual evidence of the system in operation. Screenshots were captured during testing to demonstrate robot movement, sensor data, and map construction.

In Figure 1 the robot is just placed in the environment not moving only explore the environment so when I start the mapper file it moves in the environment.

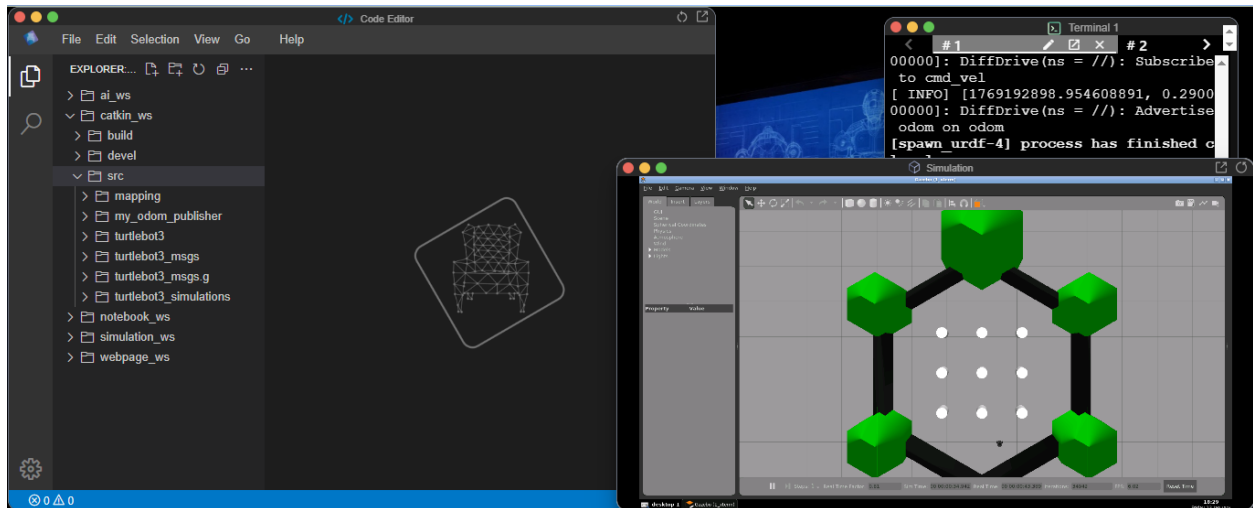


FIGURE SHOWS THE TURTLEBOT3 ROBOT POSITIONED INSIDE THE SIMULATED ENVIRONMENT BEFORE EXPLORATION BEGINS.

In this I run the rviz and grid cells is shown by using the grid data as given in the coursework.

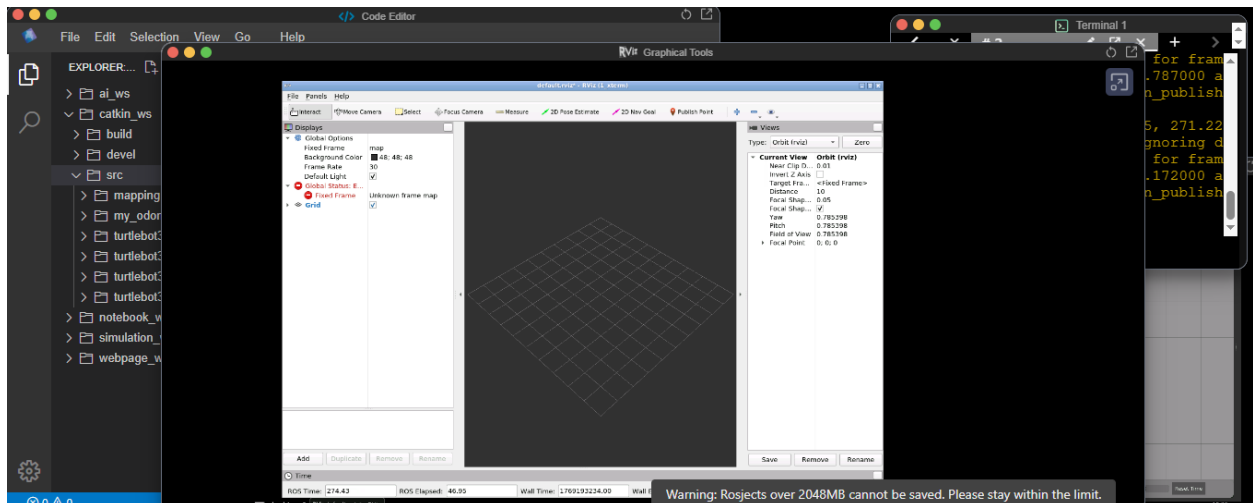


FIGURE SHOWS THE RVIZ GRID CELLS WITHOUT MAP

Figure 3 demonstrates the active mapping process. The mapper node is running, and RViz displays the generated 2D occupancy grid map. As the robot moves, sensor data is continuously updated, allowing obstacles and free spaces in the environment to be visualised in real time.

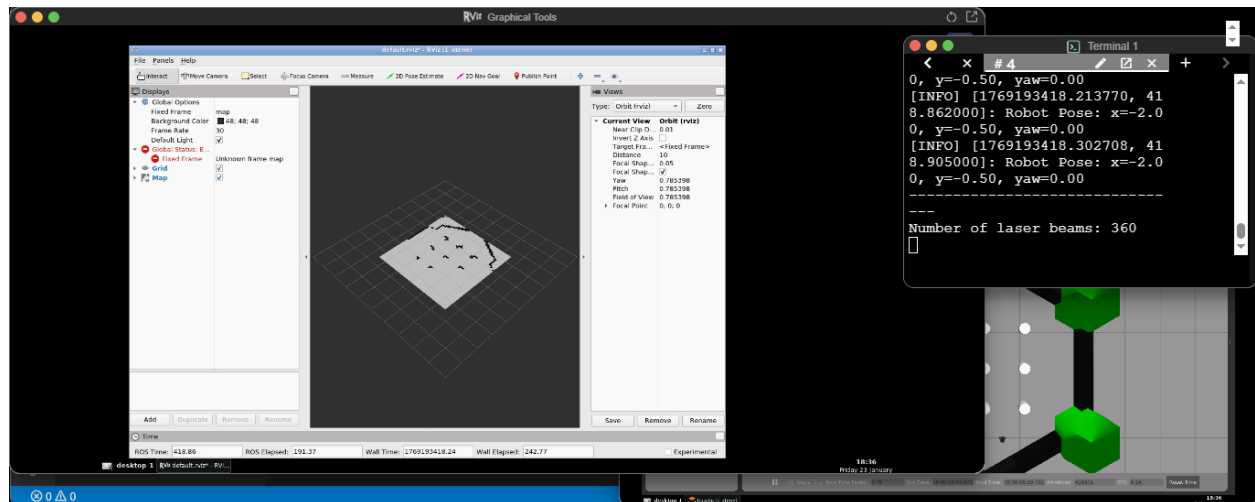


FIGURE SHOWS THAT RVIZ WITH 2D GRAPH INSIDE AND MAPPER FILE IS RUNNING ALONGSIDE

Testing and Results:

The robot was driven tele-operatively by means of a keyboard in the world. As the robot moved, laser scans from timesteps were transformed and the occupancy grid was updated.

The findings demonstrate that the system was able to produce a 2D map which closely corresponds to the simulated environment.

Conclusions:

This project is successfully implemented 2D occupancy grid mapping using TurtleBot3. Using only odometry and laser sensor data, showing understanding of mapping without referencing ROS navigation libraries.

References:

1. <https://docs.pal-robotics.com/25.01/navigation/mapping-tutorial.html>
2. https://wiki.ros.org/costmap_2d
3. <https://emanual.robotis.com/>